

Homework #1

Illya Starikov

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Theorem 1. *There does not exist a level set with more than one bottom level.*

Proof. Suppose that there exists two, different levels P and Q in the level set M (where M may contain two — P and Q — or more levels). Per our definition, P and Q are bottom levels of the level set M if there exists a level x in M where x is not above P and Q . This implies P cannot be above Q and Q cannot be above P , for a level x must not be above P and Q . However, this violates Axiom 3 (which states that P is above Q or Q is above P). This has lead us to a contradiction. $\therefore$ There does not exists a level set with more than one bottom level. $\square$